THE LOCUS WHERE A FINITELY PRESENTED MODULE IS FLAT IS A DISJOINT UNION OF RANK STRATA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The card is taken because the representability of F_flat is proved in full; the trailing sentence 'The other statements also follow from the lemma' delegates only the 'moreover' to the cited rank lemma, and that delegation is recorded here for the reader.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The closed subschemes and functors measuring the locally free rank. Lemma lemma-locally-free-rank-r-pullback, used at proof lines 1466-1468 and 1491-1499. Supplies Z_r and the functors F_r representing the rank-r locus.
- A flat module of finite presentation is finite locally free. Properties, Lemma properties-lemma-finite-locally-free, used at proof lines 1487-1489. Turns flatness into a coproduct decomposition by rank.

Source reference. Original lemma 05P8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7332–7332. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05P2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7326–7326. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $\mathcal{F}$ be an $\mathcal{O}_S$-module of finite presentation. Let $S = Z_{-1} \supset Z_0 \supset Z_1 \supset \ldots$ be as in Lemma 05P8. Set $S_r = Z_{r-1} \setminus Z_r$. Then $S' = \coprod_{r \geq 0} S_r$ represents the functor*

$$F_{flat} : Sch/S \longrightarrow Sets, \qquad T \longmapsto \begin{cases} \{*\} & \text{if } \mathcal{F}_T \text{ flat over } T \\ \emptyset & \text{otherwise} \end{cases}$$

Moreover, $\mathcal{F}|_{S_r}$ is locally free of rank r and the morphisms $S_r \to S$ and $S' \to S$ are of finite presentation.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Suppose that $g : T \to S$ is a morphism of schemes such that the pullback $\mathcal{F}_T = g^*\mathcal{F}$ is flat. Then $\mathcal{F}_T$ is a flat $\mathcal{O}_T$-module of finite presentation. Hence $\mathcal{F}_T$ is finite locally free, see Properties, Lemma 05P2. Thus $T = \coprod_{r \geq 0} T_r$, where $\mathcal{F}_T|_{T_r}$ is locally free of rank r. This implies that

$$F_{flat} = \coprod\nolimits_{r \geq 0} F_r$$

in the category of Zariski sheaves on Sch/S where F_r is as in Lemma 05P8. It follows that F_{flat} is represented by $\coprod_{r \geq 0}(Z_{r-1} \setminus Z_r)$ where Z_r is as in Lemma 05P8. The other statements also follow from the lemma. □